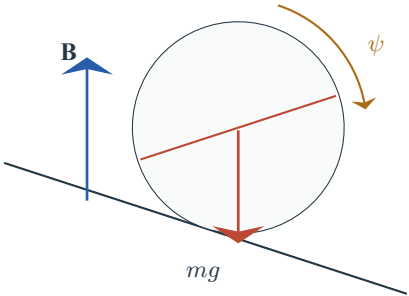
M5-166 滚动线框的平衡与稳定性



实心圆柱无滑动滚动；$x = r\psi$

$$J_{\text{eff}} = 3mr^2/2$$

$$U = \frac{2B^2r^2h^2}{L}(1-\cos\psi)^2 - mgr\sin\alpha\,\psi$$

$$U'(\psi_0) = 0, \quad U''(\psi_0) > 0$$